

CAB BOOKING SYSTEM USING MERN STACK

Project Documentation

1.Introduction

Project Title: CAB BOOKING SYSTEM

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2. Project Overview

- **Purpose:** The **Cab Booking System** is a full-stack web application developed using the **MERN Stack (MongoDB, Express.js, React.js, and Node.js)**. The purpose of this project is to provide a simple, secure, and user-friendly platform that allows users to browse available cabs, book rides, and manage their bookings online. The system also enables administrators to manage cab details efficiently through an admin interface. The project demonstrates the implementation of modern web development concepts such as RESTful APIs, user authentication, database integration, and responsive user interface design.
- **Features:** The Cab Booking System provides secure user registration and login using JWT-based authentication. Users can browse available cabs, book rides by entering pickup and drop locations along with the booking date, and view their booking history in the **My Bookings** section. The system also includes an admin module to add, update, and delete cab details. Built with the MERN Stack, it offers a responsive user interface and uses RESTful APIs for efficient communication between the frontend and backend.

3. Architecture

Frontend

The frontend of the Cab Booking System is developed using **React.js**. It consists of reusable components and pages for user registration, login, cab listing, booking, and booking management. **React Router** is used for navigation between pages, while **Axios** is used to communicate with the backend APIs. The application provides a responsive and user-friendly interface using HTML and CSS.

Backend

The backend is built using **Node.js** and **Express.js**. It follows a RESTful API architecture to handle user authentication, cab management, and booking operations. Express routes process client requests, controllers implement the business logic, and middleware is used for JWT authentication and request validation. The backend securely communicates with the MongoDB database.

Database

The application uses **MongoDB** as the database to store user, cab, and booking information. **Mongoose** is used as the Object Data Modeling (ODM) library to define schemas and perform database operations. The database supports efficient storage, retrieval, updating, and deletion of records, ensuring smooth interaction between the application and the stored data.

4. Setup Instructions

Prerequisites

The following software and tools are required to run the Cab Booking System:

- Node.js
- npm (Node Package Manager)
- MongoDB Atlas or MongoDB Community Server
- Visual Studio Code
- Git
- Web Browser (Google Chrome or Microsoft Edge)

Installation

Step 1: Clone the project repository.

git clone : <https://github.com/soumyabandlamudi/UCab.git>

Step 2: Navigate to the project directory.

```
cd CabBookingSystem
```

Step 3: Install frontend dependencies.

```
cd frontend
```

```
npm install
```

Step 4: Install backend dependencies.

```
cd ../backend
```

```
npm install
```

Step 5: Create a .env file in the backend folder and configure the environment variables.

```
PORT=5000
```

```
MONGO_URI=mongodb://localhost:27017/UCab
```

```
JWT_SECRET=your_secret_key
```

Step 6: Start the backend server.

```
npm start
```

Step 7: Start the frontend application.

```
cd ../frontend
```

```
npm start
```

The application will run locally and can be accessed through the browser.

5. Folder Structure

Client (Frontend)

The client folder is developed using **React.js** and contains the user interface of the application. It is organized into different folders for better maintainability and code reusability.

- **components/** – Contains reusable UI components such as Navbar and other common elements.
- **pages/** – Includes application pages such as Home, Login, Register, Book Cab, My Bookings, and Admin Dashboard.
- **services/** – Contains API service files for communicating with the backend.
- **styles/** – Stores CSS files used for styling the application.
- **App.js** – Main application component that manages routing.
- **main.jsx** – Entry point of the React application.

Server (Backend)

The server folder is built using **Node.js** and **Express.js** and handles the application's business logic, authentication, and database operations.

- **controllers/** – Contains the logic for handling API requests.
- **models/** – Defines MongoDB schemas using Mongoose.
- **routes/** – Contains API routes for users, cabs, and bookings.
- **middleware/** – Stores authentication middleware for protecting routes.
- **config/** – Contains database configuration files.
- **server.js** – Entry point of the backend server.
- **package.json** – Contains project dependencies and scripts.

6. Running the Application

To run the Cab Booking System locally, start both the backend and frontend servers in separate terminals.

Backend Server

Navigate to the backend folder and run:

```
cd backend
```

```
npm start
```

The backend server starts on **http://localhost:5000**.

Frontend Server

Navigate to the frontend folder and run:

```
cd frontend
```

```
npm start
```

The frontend application starts on **http://localhost:3000** and can be accessed through a web browser.

7. API Documentation

Method	Endpoint	Description	Parameters
POST	/api/auth/register	Register a new user	Name, Email, Password
POST	/api/auth/login	Authenticate user	Email, Password
GET	/api/cabs	Retrieve all available cabs	None
GET	/api/cabs/:id	Retrieve cab details by ID	Cab ID
POST	/api/bookings	Create a new cab booking	Pickup Location, Drop Location, Booking Date
GET	/api/bookings	Retrieve user bookings	JWT Token
DELETE	/api/bookings/:id	Cancel/Delete a booking	Booking ID
POST	/api/cabs (Admin)	Add a new cab	Cab Details

PUT	/api/cabs/:id (Admin)	Update cab information	Cab ID, Updated Details
DELETE	/api/cabs/:id (Admin)	Delete a cab	Cab ID

Example API Request

POST /api/auth/login

Request Body

```
{
  "email": "user@example.com",
  "password": "password123"}
```

Example Response{

```
"message": "Login successful",
"token": "<JWT_TOKEN>"
```

8. Authentication

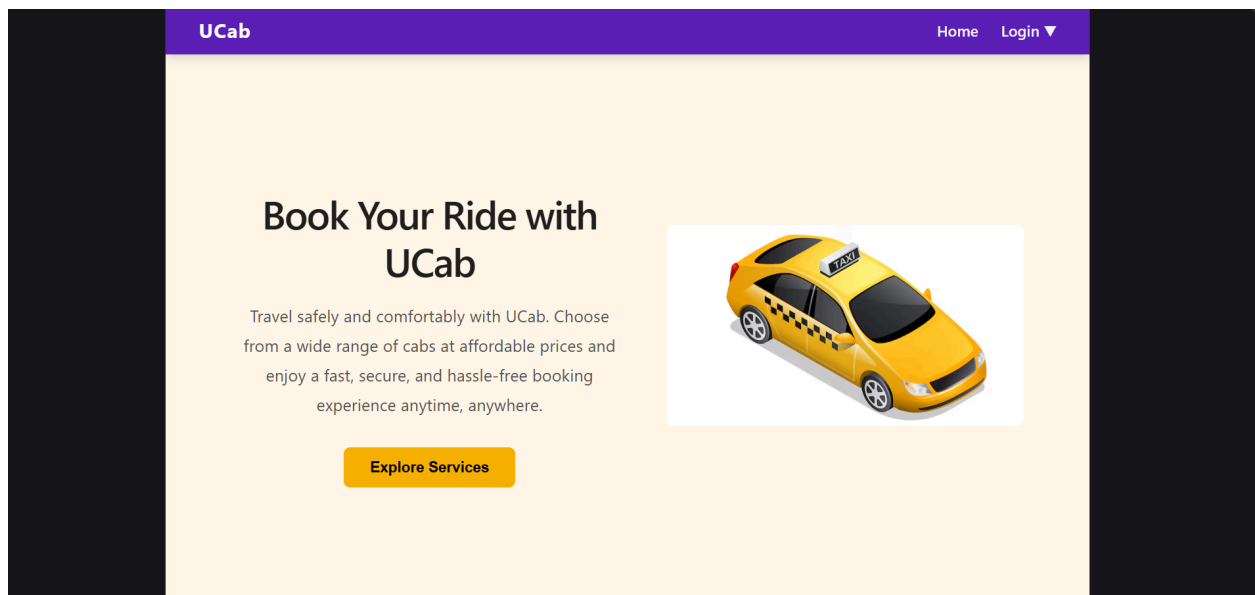
The Cab Booking System uses **JSON Web Token (JWT)** based authentication to provide secure access to the application. Users must register and log in using their email and password. After successful authentication, the server generates a JWT token, which is sent to the client and used to access protected routes.

The token is included in subsequent API requests to verify the user's identity. Authentication middleware on the backend validates the token before allowing access to protected resources such as booking management and admin operations. This approach ensures that only authenticated users can perform authorized actions while maintaining the security of the application.

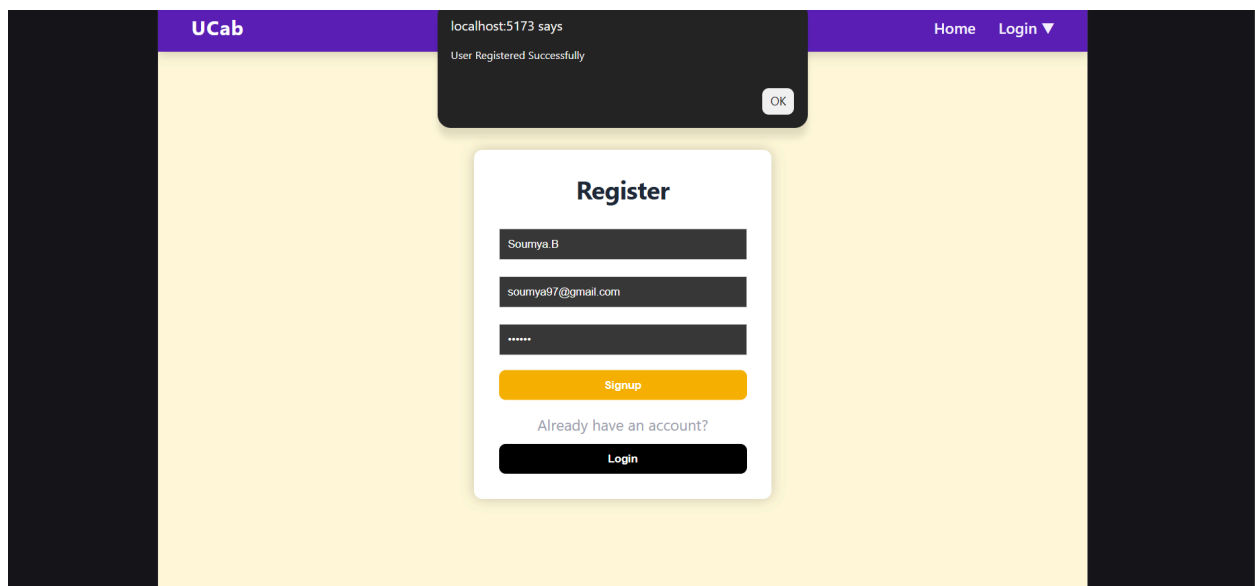
9. User Interface

The Cab Booking System provides a clean, responsive, and user-friendly interface for both users and administrators. The following screenshots demonstrate the major features of the application.

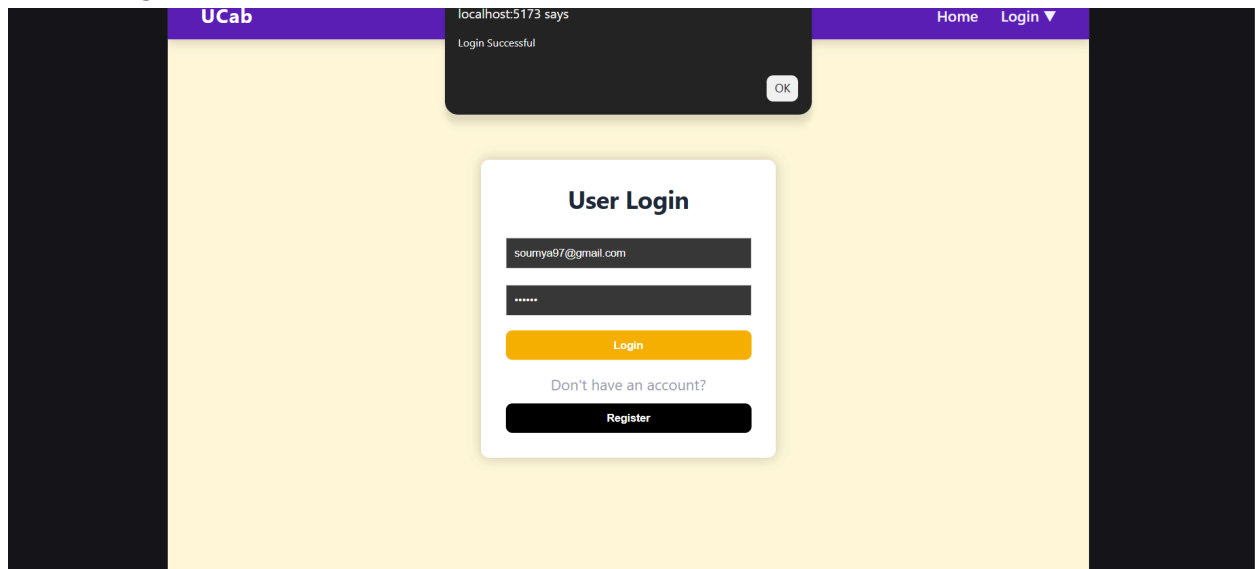
1.Homepage



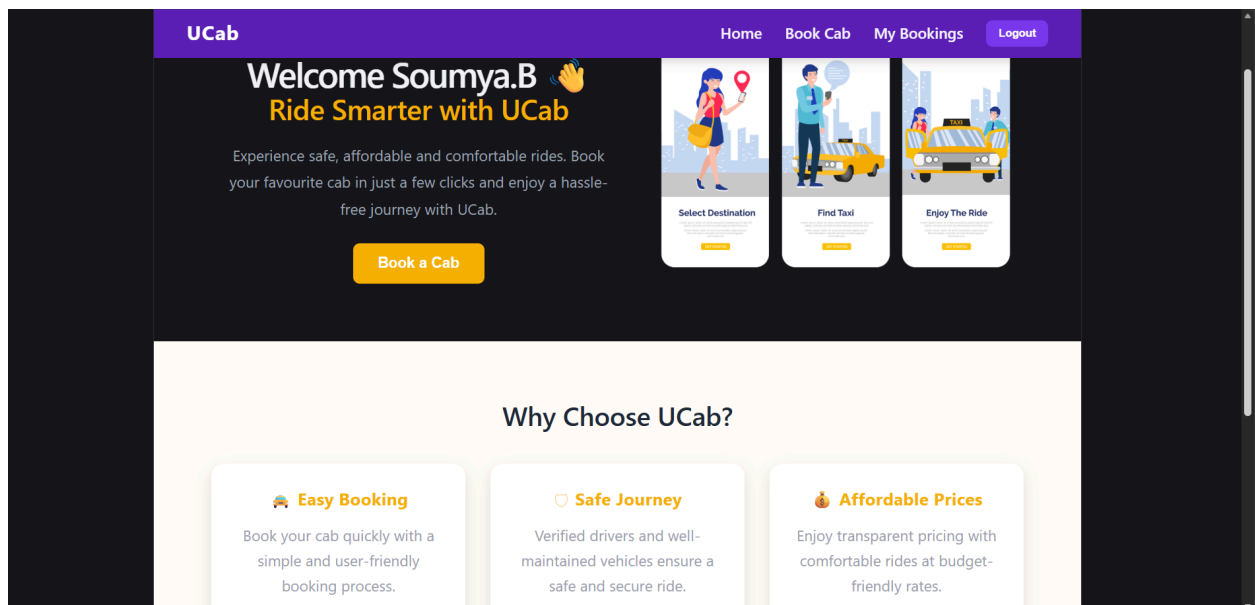
2.User SignUp

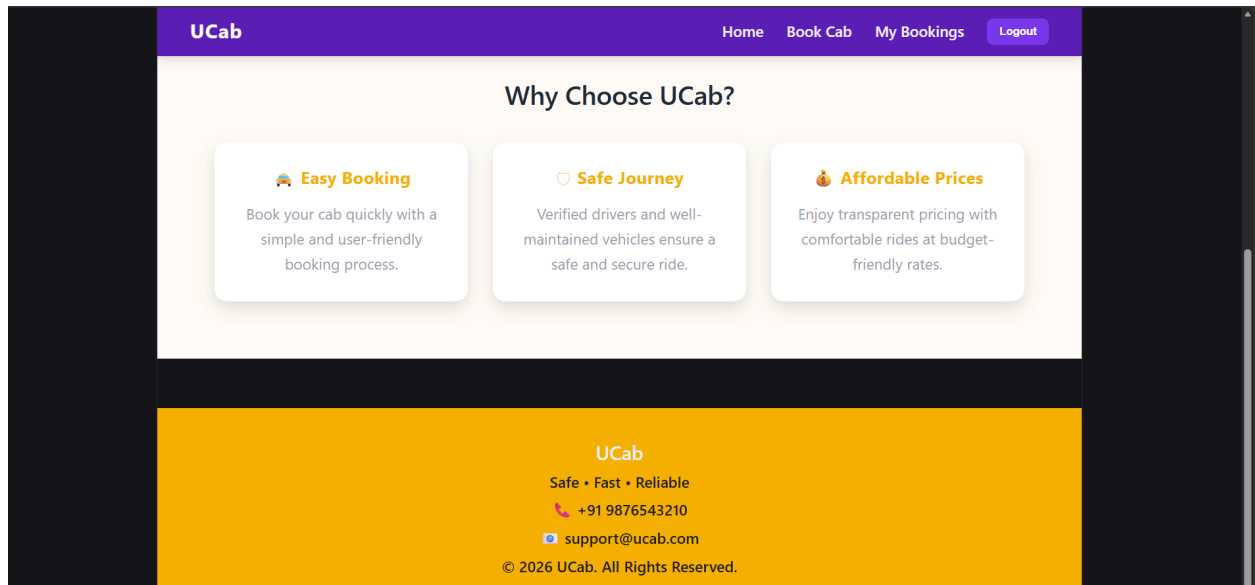


3. User Login

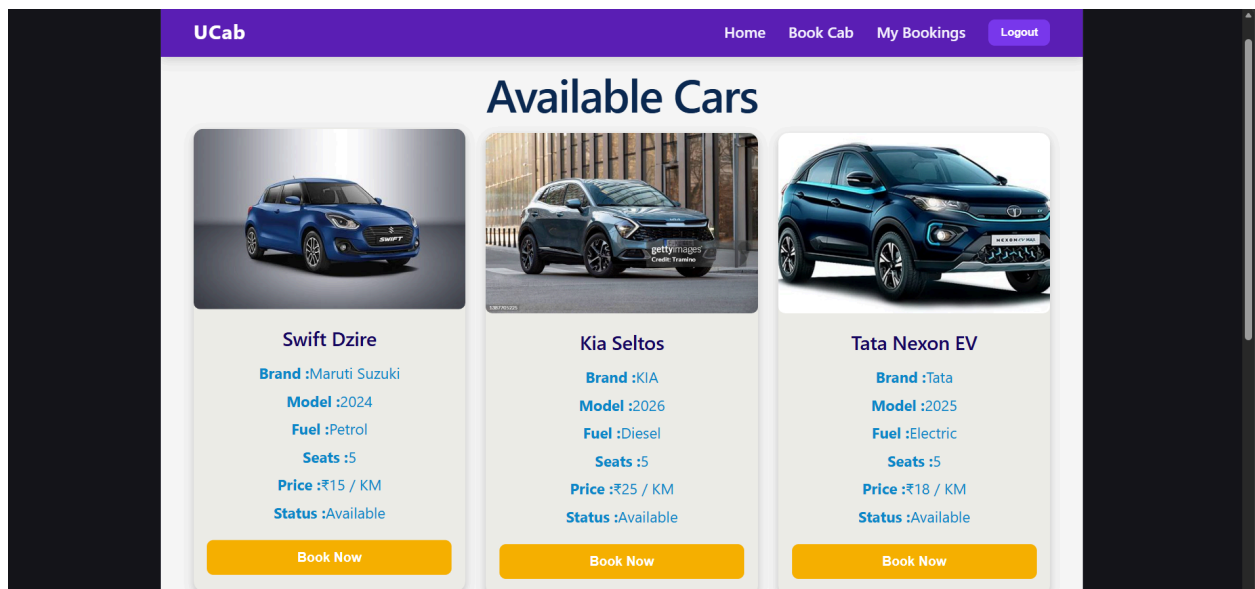


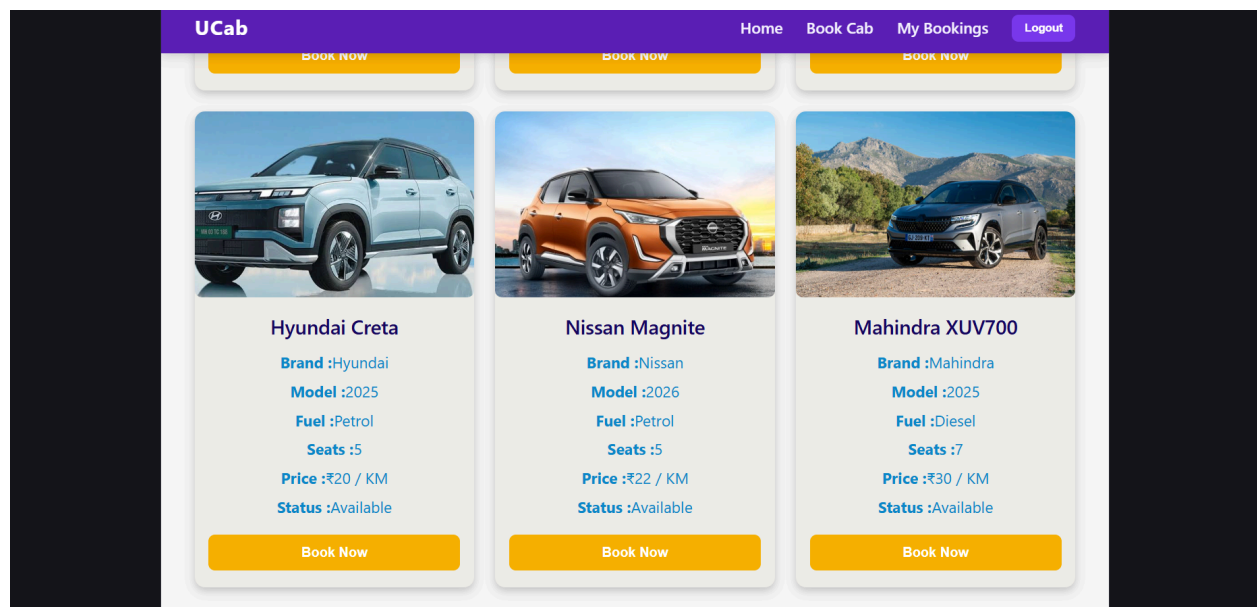
4. User Interface



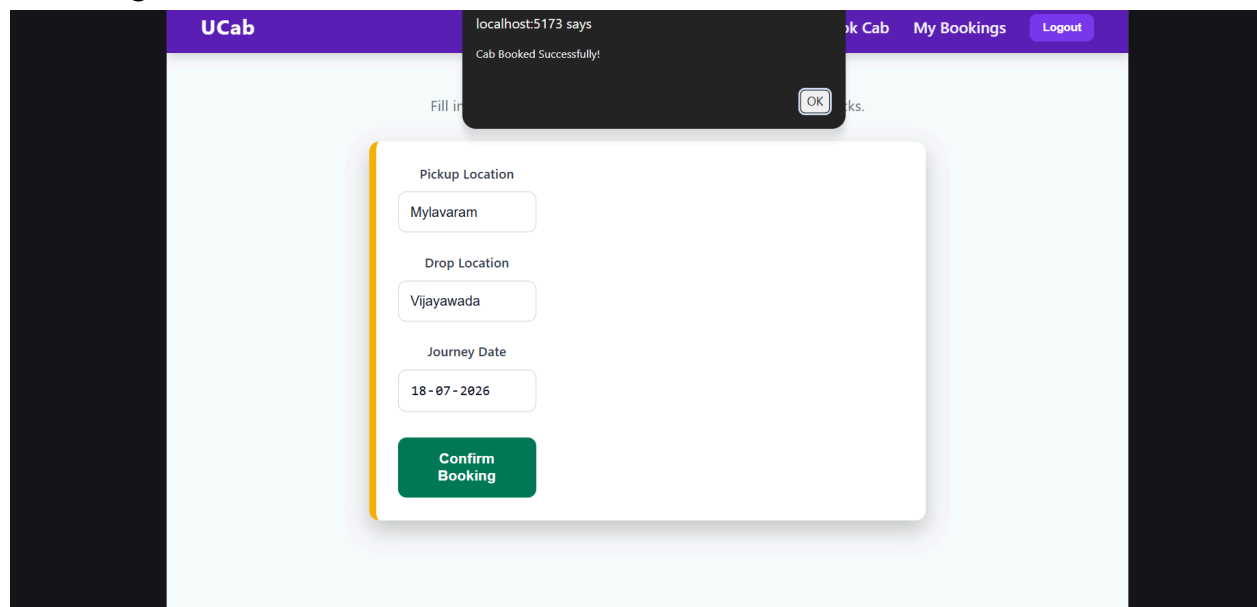


5.Book a Cab





6.Booking Form



7. MyBookings

UCab

[Home](#) [Book Cab](#) [My Bookings](#) [Logout](#)

My Bookings

Booking Date	Pickup	Drop	Car
18/7/2026	Mylavaram	Vijayawada	Swift Dzire
Brand	Seats	Price / Km	Status
Maruti Suzuki	5	₹ 15	Booked

Booking Date	Pickup	Drop	Car
21/7/2026	Vijayawada	Vissannapeta	Tata Nexon EV
Brand	Seats	Price / Km	Status
Tata	5	₹ 18	Booked

8.User Logout

UCab

localhost:5173 says
Logged Out Successfully

OK

[Book Cab](#) [My Bookings](#) [Logout](#)

My Bookings

Booking Date	Pickup	Drop	Car
18/7/2026	Mylavaram	Vijayawada	Swift Dzire
Brand	Seats	Price / Km	Status
Maruti Suzuki	5	₹ 15	Booked

Booking Date	Pickup	Drop	Car
21/7/2026	Vijayawada	Vissannapeta	Tata Nexon EV
Brand	Seats	Price / Km	Status
Tata	5	₹ 18	Booked

9.Admin Login

UCab

localhost:5173 says
Admin Login Successful!
OK

Home Login ▾

Admin Login

admin@gmail.com

Login

10.Admin Dashboard

UCab

Dashboard Logout

Admin Dashboard

Welcome back, Administrator. Manage users, cabs and bookings from one place.



Add Cab

Register a new cab into the system.



Manage Cabs

View, edit and delete available cabs.



Users

View all registered customers.



Bookings

Track and manage customer bookings.

11.Add a Cab

UCab

localhost:5173 says
Car Added Successfully

DashboardLogout

OK

Add New Car

Hyundai Creta

Hyundai

2026

25

5

1.2000

Choose FileNissan.jpg

Add Car

12.Manage Cars

UCab

DashboardLogout

Manage Cars

Image	Name	Brand	Model	Fuel	Seats	Price/Km	Actions
	Swift Dzire	Maruti Suzuki	2024	Petrol	5	₹ 15	<button>Edit</button> <button>Delete</button>
	Kia Seltos	KIA	2026	Diesel	5	₹ 25	<button>Edit</button> <button>Delete</button>
	Tata Nexon EV	Tata	2025	Electric	5	₹ 18	<button>Edit</button> <button>Delete</button>
	Hyundai Creta	Hyundai	2025	Petrol	5	₹ 20	<button>Edit</button> <button>Delete</button>
	Nissan Magnite	Nissan	2026	Petrol	5	₹ 22	<button>Edit</button> <button>Delete</button>
	Mahindra XUV700	Mahindra	2025	Diesel	7	₹ 30	<button>Edit</button> <button>Delete</button>

13.Update Cars

UCab

localhost:5173 says
Car Updated Successfully

DashboardLogout

OK

Swift Dzire

Maruti Suzuki

2024

Petrol

5

25

Choose FileNo file chosen

Update Car

14.Delete a Car

UCab	localhost:5173 says Car Deleted Successfully						Dashboard	Logout
	Kia Seltos						₹ 25	EditDelete
	Tata Nexon EV						₹ 18	EditDelete
	Hyundai Creta	Hyundai	2025	Petrol	5		₹ 20	EditDelete
	Nissan Magnite	Nissan	2026	Petrol	5		₹ 22	EditDelete
	Mahindra XUV700	Mahindra	2025	Diesel	7		₹ 30	EditDelete
	Hyundai Creta	Hyundai	2026	Diesel	5		₹ 25	EditDelete

15.Registered Users

UCab		Dashboard	Logout
Registered Users			
Name	Email		
Soumya	soumya@gmail.com		
soumya	soumyabandlamudi06@gmail.com		
Sujani	sujani17@gmail.com		
Soumya	soumya06@gmail.com		
Suresh	suresh6@gmail.com		
Demo	demo@gmail.com		
Sample	sample@gmail.com		
Soumya	soumya01@gmail.com		
Soumya.B	soumya97@gmail.com		

16.All Bookings

UCab

DashboardLogout

All Bookings

User	Car	Pickup	Drop	Date	Status
Soumya	N/A	Vijayawada	Guntur	15/7/2026	Cancelled
soumya	Swift Dzire	vissannapeta	mylavaram	13/7/2026	Booked
soumya	Swift Dzire	vissannapeta	mylavaram	13/7/2026	Booked
soumya	Swift Dzire	vissannapeta	mylavaram	12/7/2026	Booked
soumya	Swift Dzire	vissannapeta	mylavaram	12/7/2026	Booked
soumya	Swift Dzire	vissannapeta	mylavaram	14/7/2026	Booked
soumya	Swift Dzire	vissannapeta	mylavaram	13/7/2026	Booked
soumya	Swift Dzire	vissannapeta	mylavaram	13/7/2026	Booked
Sujani	Swift Dzire	Mylavaram	Vissannapeta	15/7/2026	Booked
Sujani	Swift Dzire	VSPT	VZA	16/7/2026	Booked
Suresh	Swift Dzire	Vijayawada	Banglore	16/7/2026	Booked

17.Admin Logout

UCab

localhost:5173 says
Logged Out Successfully

Dashboard Logout

OK

User	Car	Pickup	Drop	Date	Status
Soumya	N/A	Vijayawada	Guntur	15/7/2026	Cancelled
soumya	Swift Dzire	vissannapeta	mylavaram	13/7/2026	Booked
soumya	Swift Dzire	vissannapeta	mylavaram	13/7/2026	Booked
soumya	Swift Dzire	vissannapeta	mylavaram	12/7/2026	Booked
soumya	Swift Dzire	vissannapeta	mylavaram	12/7/2026	Booked
soumya	Swift Dzire	vissannapeta	mylavaram	14/7/2026	Booked
soumya	Swift Dzire	vissannapeta	mylavaram	13/7/2026	Booked
soumya	Swift Dzire	vissannapeta	mylavaram	13/7/2026	Booked
Sujani	Swift Dzire	Mylavaram	Vissannapeta	15/7/2026	Booked
Sujani	Swift Dzire	VSPT	VZA	16/7/2026	Booked
Suresh	Swift Dzire	Vijayawada	Banglore	16/7/2026	Booked

10.Testing

The application was tested using manual functional testing to verify the correctness of all major features. API endpoints were tested using Postman, while the frontend was tested in Google Chrome. User authentication, cab booking, booking management, and CRUD operations were verified to ensure the application functions as expected.

11.Demo video:

https://drive.google.com/file/d/1P8MR6D7Tk0W0Q_6DjKHqiOm4latqWg8r/view?usp=drive_link

12. Known Issues:

- The application does not support online payment integration.
- Real-time cab tracking is not available.
- Email or SMS notifications are not implemented.
- The application currently supports basic booking management features only.

13. Future Enhancements

- Integrate an online payment gateway.
- Add real-time cab tracking using Google Maps.
- Implement email and SMS booking notifications.
- Introduce ride history and booking analytics.
- Add user ratings and reviews for drivers.
- Develop a mobile application for Android and iOS.
- Implement advanced search and filtering options for cabs.